

Samsung Securities

2025 Outlook Semiconductor MPE



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Semiconductor MPE investment strategy

Equipment

- 1) Key Idea: The scalability through R&D needs to be better reflected in valuation
- 2) Preference: HPSP, Park Systems

Materials/Parts

- 1) Key Idea: Price reflected utilization rate adjustment. Gradual accumulation strategy remains valid.
- 2) Preference: Soulbrain, Leeno Industrial

OSAT

- 1) Key Idea: Outsourcing volume of memory back-end manufacturing increases, Finding a new growth foundation
- 2) Preference: Hana Micron

PART 01

Capital investment: Adjusting high expectations



Equipment: How to approach ahead

Stock price drivers

- 1) Profit drivers: Client capital investment and utilization rate
- 2) Valuation drivers: Position in the semiconductor cycle, response to technological changes

Scale of capital investment

A downward adjustment in capital investment is expected in 2025. While the market may grow slightly, growth rates could underperform expectations due to rising equipment costs and increased technological complexity.

Trends in capital investment

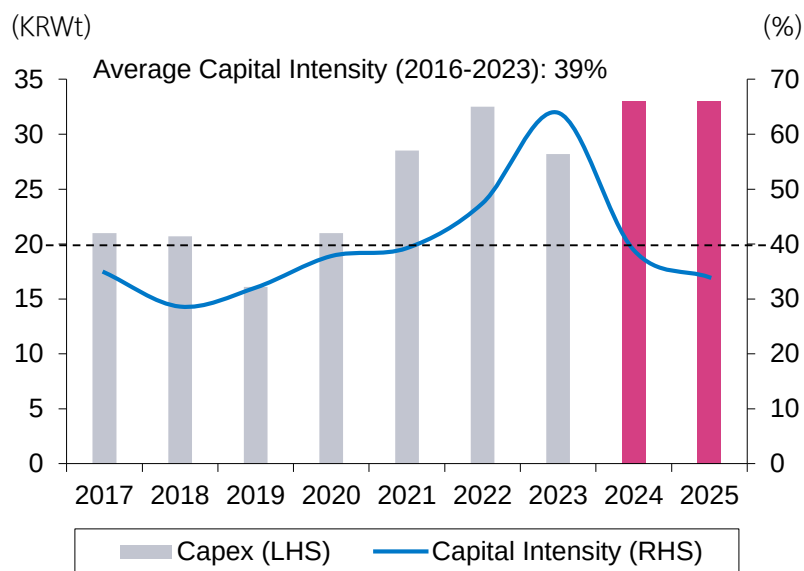
Capital investment will be polarized in 2025, with reduced spending on conventional semiconductors and increased focus on advanced semiconductor (DDR5, AI chips) production.

Current position

Stock price reflected the impact of downcycle, presenting an opportunity for low-price entry. R&D capability expansion is needed to outperform in this recovery cycle.

2025 Samsung Memory Capex: Technology investment

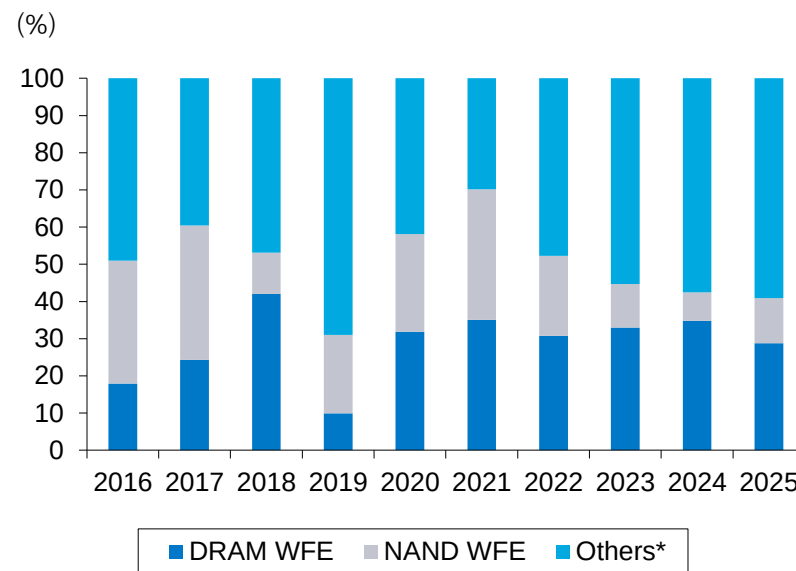
Samsung Memory: Capex trends



- 2024 Capex: DRAM 1b tech migration, HBM back-end investment, early execution of infra investment (P4L), around 33 KRWt in total.
- 2025 Capex: Investment for short-term production will decrease, with focusing on R&D to restore core technological competitiveness

Source: Company data, Samsung Securities estimates

Samsung Memory: Capex mix



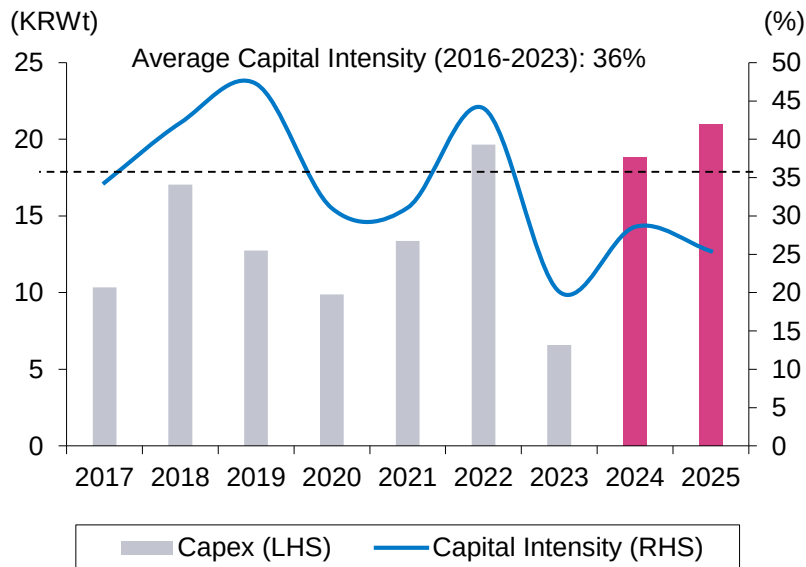
- DRAM: Equipment investment is expected to increase slightly. Mainly on DRAM 1b tech migration (for high-value DDR5 demand). New EUV purchases are expected to decrease.
- NAND: Focusing on V9 migration and V10 R&D investment.

Note: *Infrastructure, R&D

Source: Samsung Securities estimates

2025 SK Hynix Capex: Risk Hedging

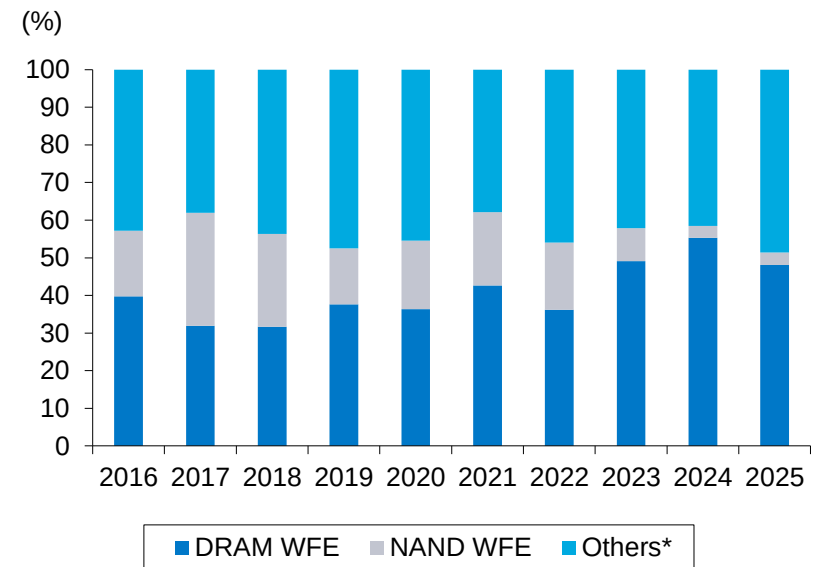
SK Hynix: Capex trends



- 2024 Capex: Focused on HBM demand (DRAM 1b, TSV). NAND equipment investment minimized from last year.
- 2025 Capex: Conservative stance to reduce oversupply risk and improve financial stability.

Source: Company data, Samsung Securities estimates

SK Hynix: Capex mix



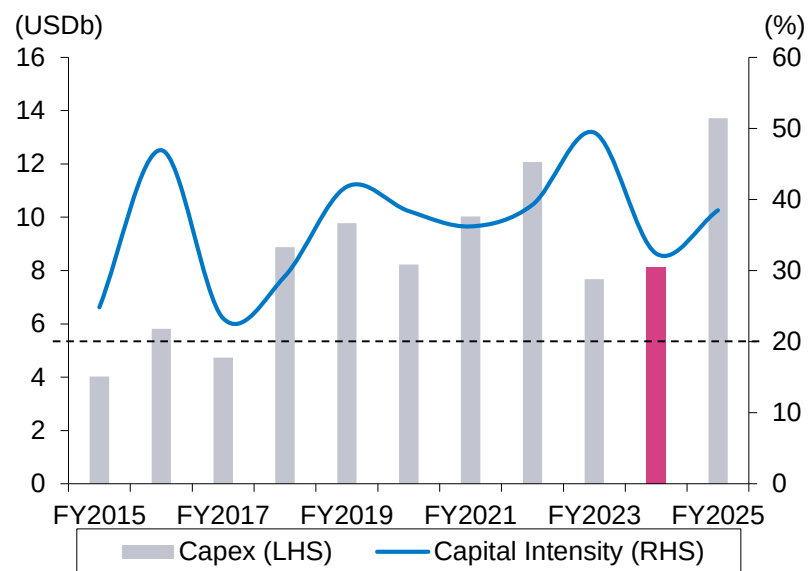
- DRAM: Investment will be concentrated in 1H (DRAM 1b, TSV), with flexible adjustments in 2H (for 2026 production) based on market conditions.
- NAND: Focusing on advanced nodes (238/321L). Legacy capacity(~176L) will decrease y-y for production adjustment

Note: *Infrastructure, R&D

Source: Samsung Securities estimates

2025 Micron Technology Capex: Infra investment expands

Micron Technology: Capex trends

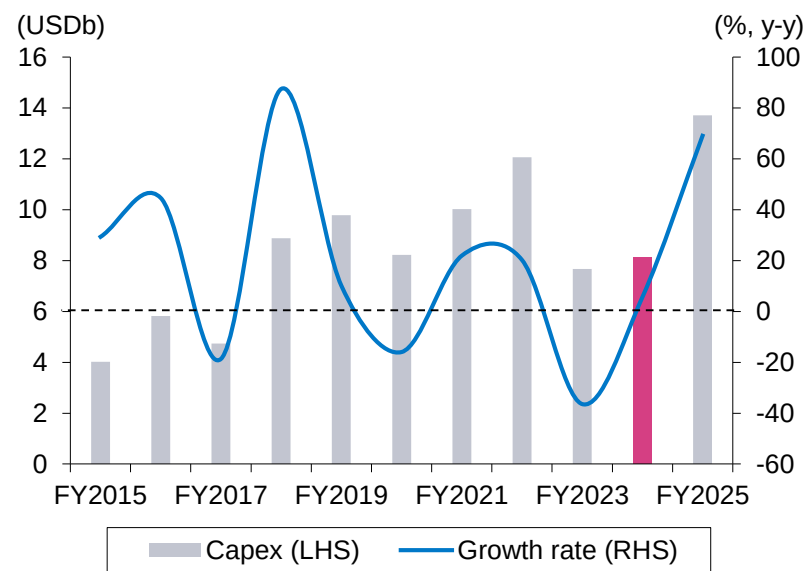


- FY2024: Focused on DRAM 1b migration for MP of DDR5 & HBM3e. Capital Intensity recorded average upturn period (low 30%).
- FY2025: Absolute Capex is expected to highly increase, mainly by increase of infra investment.

Note: FY ends August

Source: Micron Technology, Bloomberg, Samsung Securities

Micron Technology: Capex trends

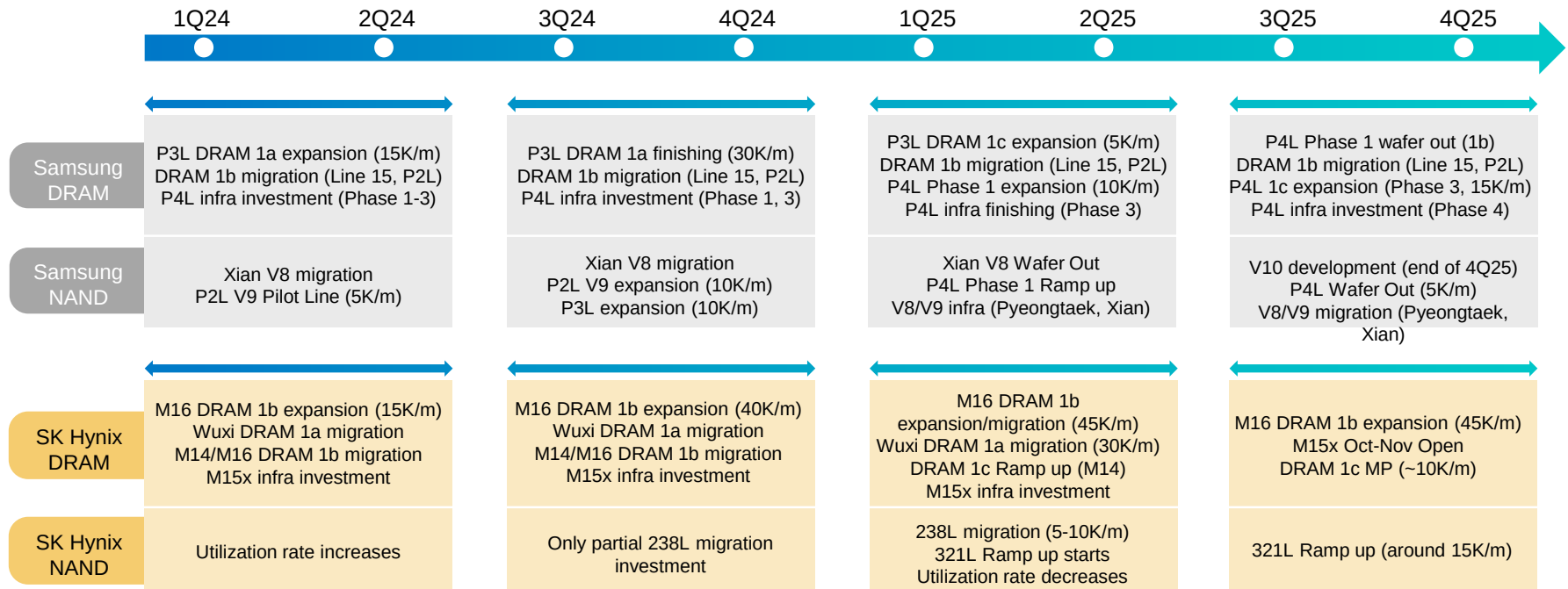


- The increase in Capex, mainly for infrastructure, may be seen as overinvestment, requiring closer observation.
- Greenfield Fab construction and HBM Capex (DRAM 1b migration) will be constitute majority of FY2025 capital investment

Note: FY ends August

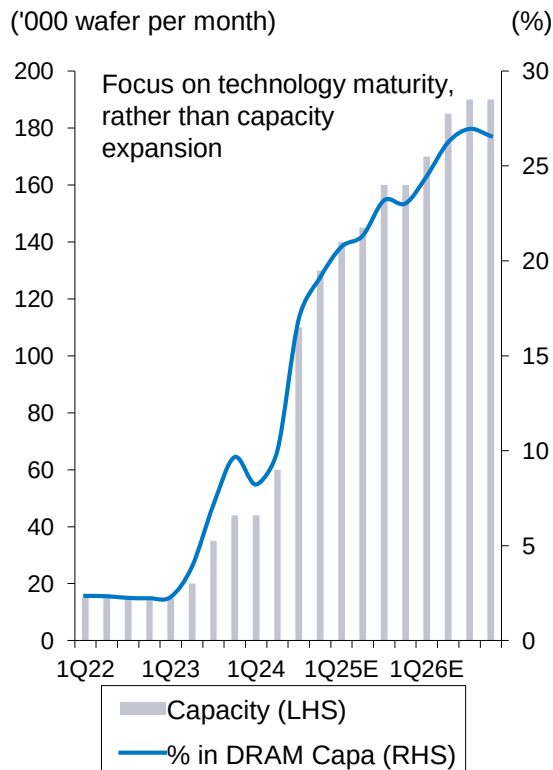
Source: Micron Technology, Bloomberg, Samsung Securities

2025 Korean memory: Realization of capital investment



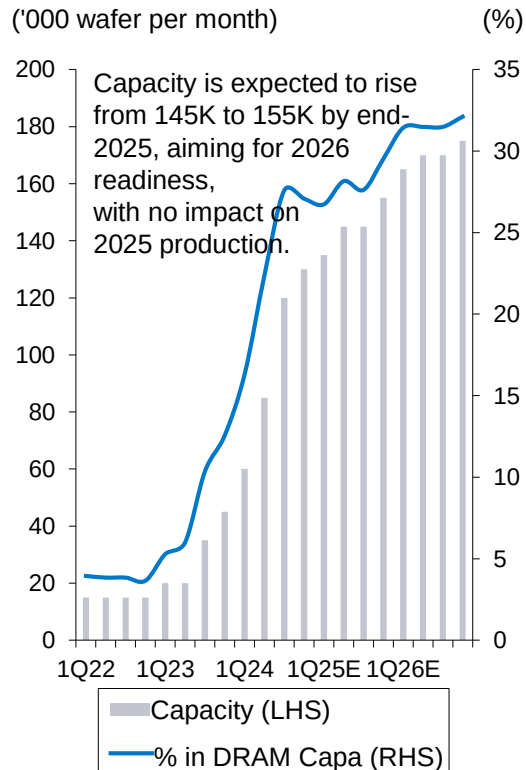
HBM back-end: Yield improvement and tech migration, rather than expansion

Samsung: HBM Capacity



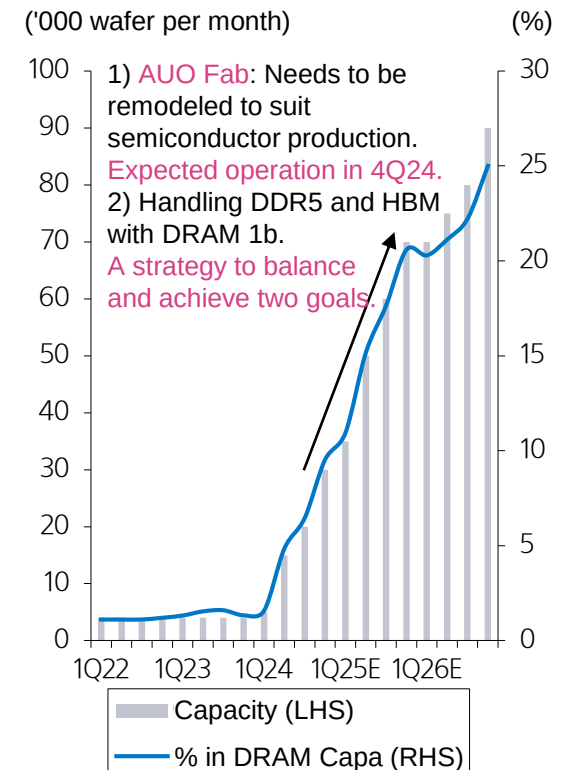
Source: Samsung Securities estimates

SK Hynix: HBM Capacity



Source: Samsung Securities estimates

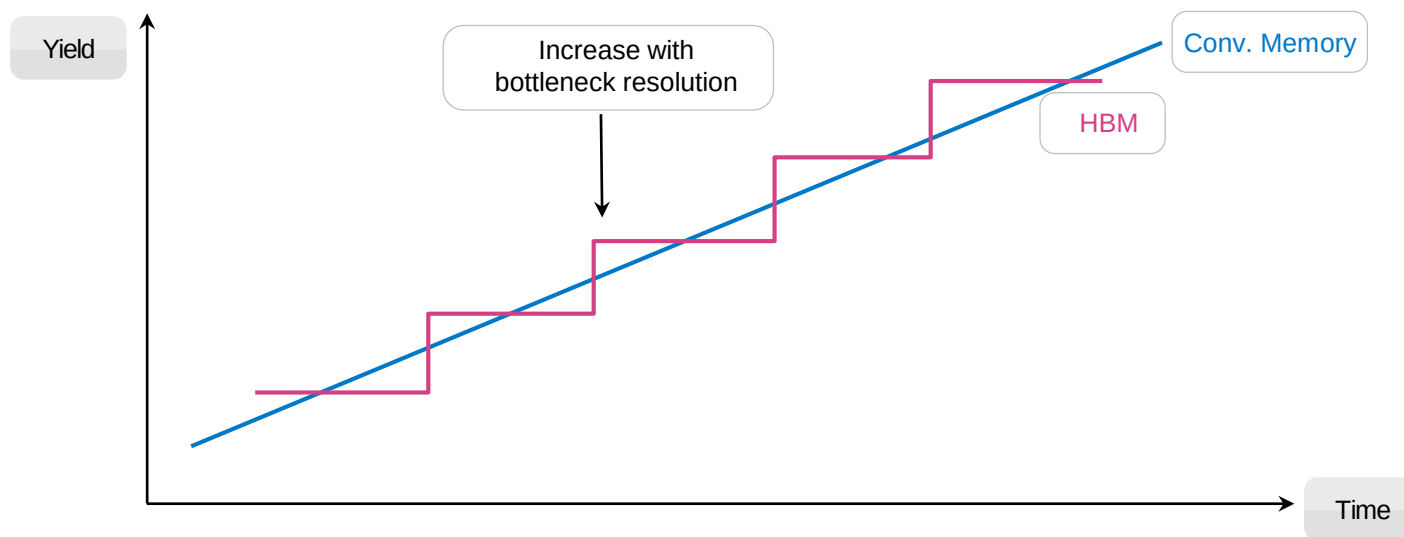
Micron: HBM Capacity



Source: Samsung Securities estimates

Why SK Hynix expands year-end capacity further?

HBM vs Conventional DRAM



- We raise our end-2025 HBM capacity forecast for SK Hynix from 145K to 155-160K. Base Case is 155K.
- Maximizing sales in 2025 is unlikely. HBM4 transition is expected in 2H, with production expansion in 1H26. HBM4 transition may reduce net die and increase yield challenges. We believe this decision is to minimize production risks in 1H26.

Source: Samsung Securities

When will the rebound timing for traditional Cycle stock be?

Key Takeaway 1

Share price preemptively reflects the downcycle. In usual cases, share price is bottomed out when share price falls by 50-60% from the previous peak, share price. We believe the current share price reflects the downcycle enough.

Key Takeaway 2

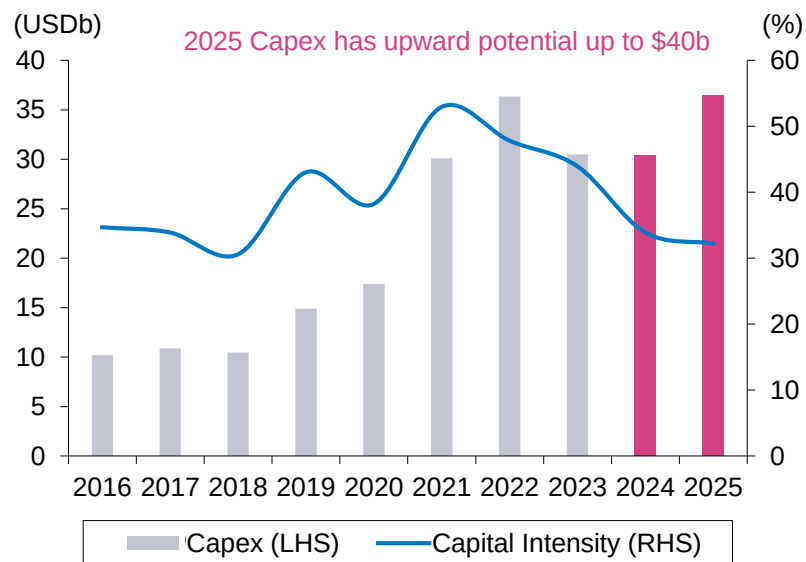
1) Front-end: Capex will be invested mainly on DRAM 1b tech migration to meet the demand for HBM3e and high-density DDR5. For the investment efficiency, the portion of tech migration gets expanded. We believe investment on DRAM 1c to be limited.
2) Back-end: HBM Capacity expansion should be less than expectation. Chipmakers focus on yield improvement and tech migration.

Key Takeaway 3

1) To outperform the rivals in recovery cycle, chipmakers should focus on R&D and AI exposure.
2) The change of demand structure is accompanied by technological change. To meet the demand on time, R&D reinforcement is essential.

2025 non-memory Capex: TSMC is only good

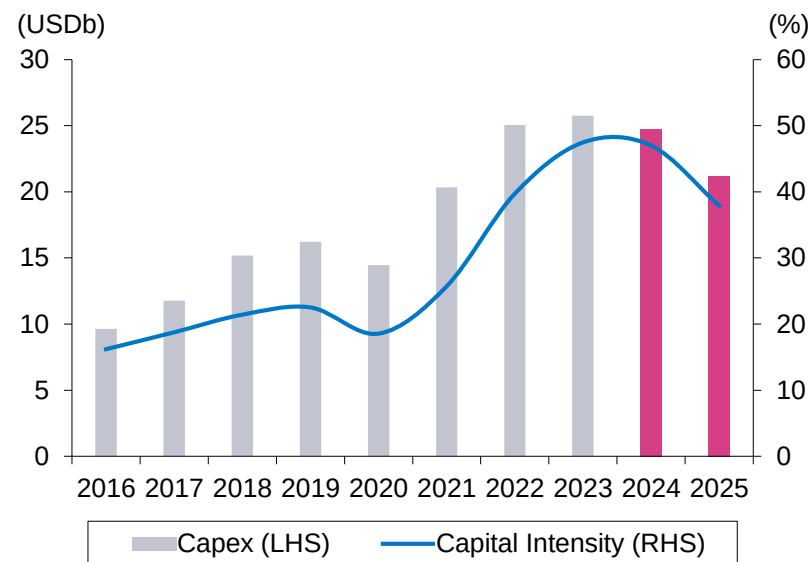
TSMC: Capex trends



- Front-end: Demand is shifting to 3nm (Mobile), requiring additional 3nm expansion. Technology investment is expected to start 2nm in 2H25.
- Back-end: Strong demand for CoWoS continues. The expansion target for end of 2025 is estimated to be raised. (Old: 60K, New: 80K)

Source: TSMC, Bloomberg, Samsung Securities

Intel: Capex trends

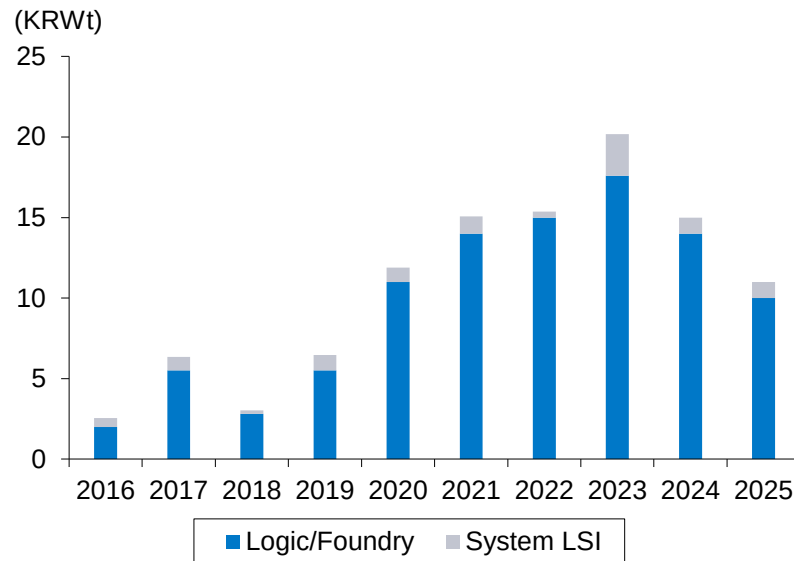


- Amid reduced competitiveness, workforce reduction continues, limiting the capacity to utilize Capex.
- Investment will focus on R&D to narrow the technology gap with TSMC, with limited execution.

Source: Intel, Bloomberg, Samsung Securities

Samsung Foundry: Reduction in equipment, increase in technology investment share

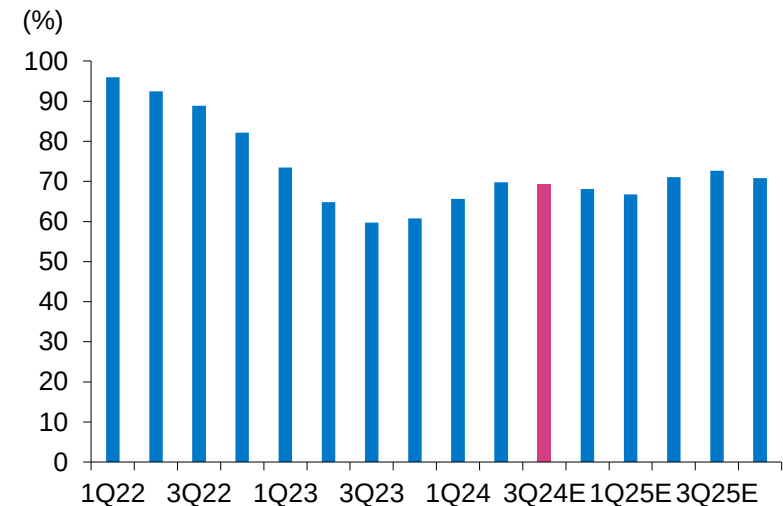
Samsung: non-memory Capex trends



- Until 2023: Shell First strategy. To secure sufficient capacity through large-scale investment and drive customer demand.
- From 2025: Capacity is sufficiently secured, with most Capex focused on enhancing technology to meet customer demand timely.

Source: Samsung Securities estimates

Samsung Foundry: utilization trends



- Little incentive for expansion, with ample space in existing plants, the focus will shift to 2/3nm technology investment and R&D.
- With rising wafer costs at TSMC, customers' diversification demand grows. Focus on driving orders through enhanced technology.

Source: Samsung Securities estimates

Equipment investment growth seen only at TSMC

TSMC

- 1) Front-end: 3nm capacity expansions and 2nm investments. Demand shifts to TSMC amid competitors' struggles.
- 2) Back-end: CoWoS capacity expansions target upgrades (Old: 60K, New: 80K)

Samsung Foundry

- 1) Front-end: No need for additional expansion. New fabs (P4L and Taylor) schedule delays. Focus on expanding competitiveness (2/3nm) in 2025
- 2) Back-end: Strengthening AVP is key to securing AI chips orders.

Intel

- 1) Front-end: Limited Capex utilization while workforce reducing. Investment is expected to focus more on technology rather than expansion.
- 2) Back-end: Focusing on enhancing MP capability of EMIB, Foveros.

PART 02

Changes in semiconductor technology



How will semiconductor technology evolve?

DRAM

DRAM 1c is expected to emerge around 1H25. The adoption of High-K Metal Gate is expected to expand further to create faster DRAM (Mobile to PC, Server)

HBM

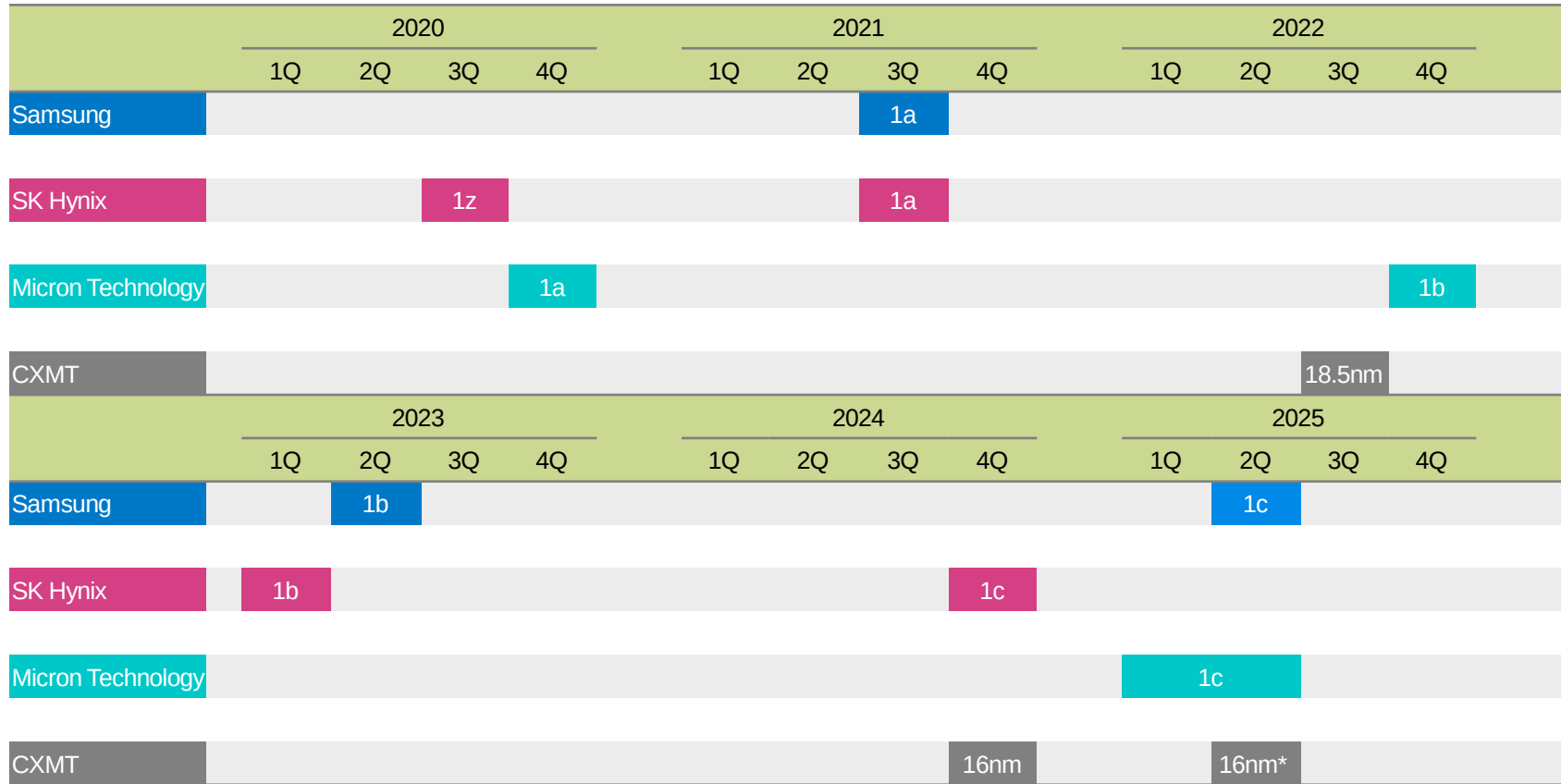
HBM4 is expected to emerge, with initial shipments anticipated around 4Q25
→ I/O counts to increase from 1,024 to 2,048 (to expand bandwidth)

NAND

Through Triple Stacking, the number of layers is expected to exceed 300 layers. Changes in Cell-Peri structure. Hybrid Bonding (Wafer to Wafer) penetration expands

DRAM: DRAM 1c emerges in 1H25

DRAM: Tech Migration Roadmap (2020-2025, by MP initiation)



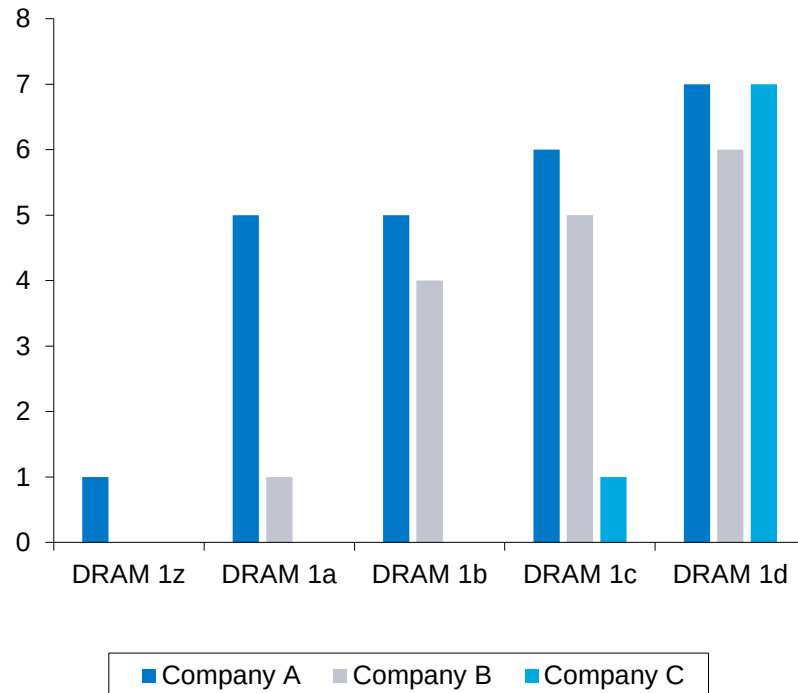
Note: *Revision version with High-K Metal Gate (Gen 4B)

Source: Samsung Securities estimates

How will DRAM technology evolve?

EUV layers by tech node

(EUV Layer)



Source: Samsung Securities estimates

Changes in processes

Process	Changes
Lithography	EUV steps increase
Etching	Limited Etching steps increase; rising specs drive higher equipment costs.
Deposition	ALD steps increase
Process Control	Total steps expected to increase for defect prevention (AFM penetration increasing, growth of Dark Field equipment, etc.)
Others	1) Laser annealing/high-pressure annealing steps increase 2) High-K Metal Gate adoption expands

Source: Samsung Securities estimates

HBM, What is the future technology roadmap?

HBM roadmap

	2013	2014	2015	2016	2017	2018	2019
Samsung			HBM2			HBM2 Aquabolt	
SK Hynix	HBM		HBM2 (2Gb)			HBM2 (8Gb)	
Micron Technology							
	2020	2021	2022	2023	2024	2025	2026
Samsung	HBM2E	HBM-PIM		HBM3	HBM3E (24Gb)	HBM4 12-hi	HBM4 16-hi
SK Hynix	HBM2 (16Gb)		HBM3 (16Gb)		HBM3E (24Gb)	HBM4 12-hi	HBM4 16-hi
Micron Technology		HBM2E			HBM3E (24Gb)	HBM4 12-hi	HBM4 16-hi

Source: Samsung Securities estimates

Changes in HBM(front-end): HBM4, Foundry processes in Logic Die

Changes in production methods by HBM generation (Front-end)

Core Die	HBM2e	HBM3	HBM3e	HBM4	HBM4e
Samsung	DRAM 1y	DRAM 1z	DRAM 1a	DRAM 1c	DRAM 1c
SK Hynix	DRAM 1y	DRAM 1z	DRAM 1b	DRAM 1b	DRAM 1c
Micron Technology	DRAM 1z	-	DRAM 1b	DRAM 1b	DRAM 1c
Logic Die	HBM2e	HBM3	HBM3e	HBM4	HBM4e
Samsung	DRAM 1y	DRAM 1z	DRAM 1a	SF4 (4nm)	SF2 (2nm)
SK Hynix	DRAM 1y	DRAM 1z	DRAM 1b	TSMC 3/5nm	TSMC 3nm
Micron Technology	DRAM 1z	-	DRAM 1b	DRAM 1b	TSMC 3nm

- HBM's key differentiation from conventional DRAM is bandwidth, with further expansion anticipated in HBM4 (I/O counts: HBM3e 1,024, HBM4 2,048)
- Adoption of foundry processes within Logic Die: Logic Die is responsible for the memory controller function of Core Die. Contributes to high-speed data processing by connecting HBM and GPUs through PHY. Foundry processes enhance memory controller functionality and add advanced logic functions, enabling customized semiconductor products.

Source: Samsung Securities estimates

Bonding in HBM: Current methods will persist for a while

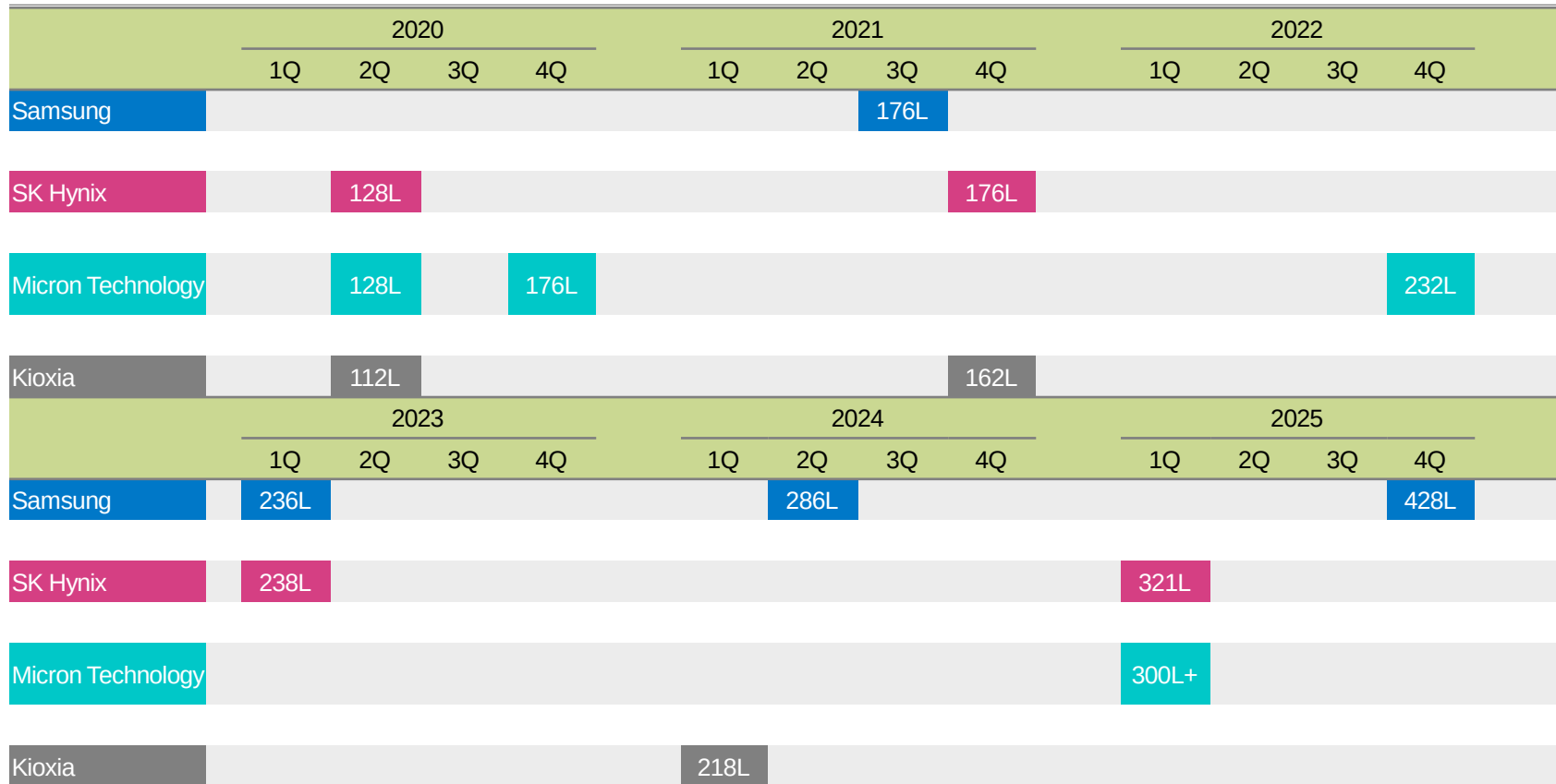
Bonding technology across HBM generations

	HBM3 8-hi	HBM3 12-hi	HBM3e 8-hi	HBM3e 12-hi	HBM4 12-hi	HBM4 16-hi	HBM4e 16-hi	HBM4e 20-hi
Samsung	TC-NCF	TC-NCF	TC-NCF	TC-NCF	TC-NCF	TC-NCF	TC-NCF	TC-NCF
SK Hynix	MR-MUF	MR-MUF	MR-MUF	MR-MUF	MR-MUF	MR-MUF	MR-MUF	MR-MUF
Micron Technology	TC-NCF	TC-NCF	TC-NCF	TC-NCF	TC-NCF	TC-NCF OR MR-MUF	TC-NCF	TC-NCF

Source: Samsung Securities estimates

NAND: Triple Stacking + Hybrid Bonding

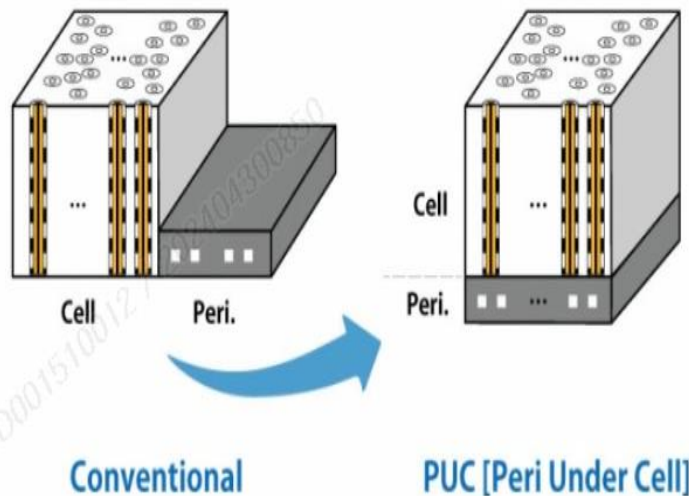
Tech Migration Roadmap in NAND industry (2020-2025, by MP initiation)



Source: Samsung Securities estimates

Hybrid Bonding (Wafer to Wafer Bonding) in NAND?

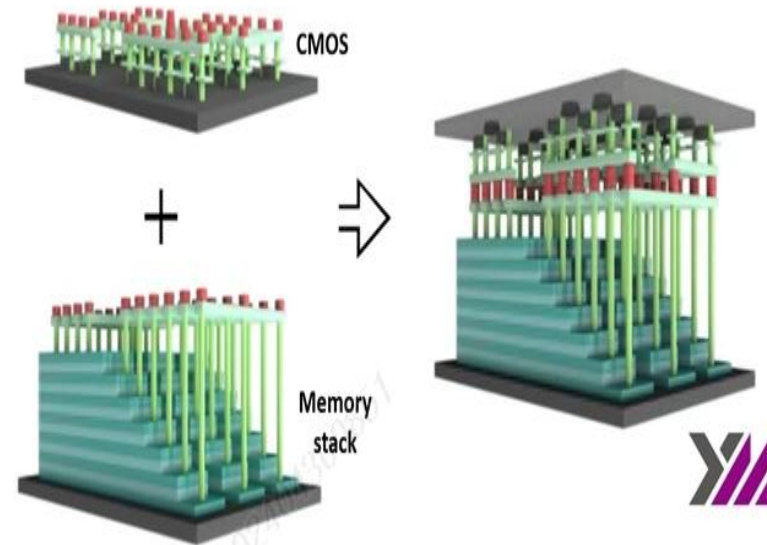
Peri Under Cell (PUC) structure



- Placing the Peri under Cell, reducing chip size and increasing the usable area of the Cell
- CTF has developed process by applying this technology starting from 96-176 layers (with PUC structure adoption timing varying by company).

Source: Industry data, Samsung Securities

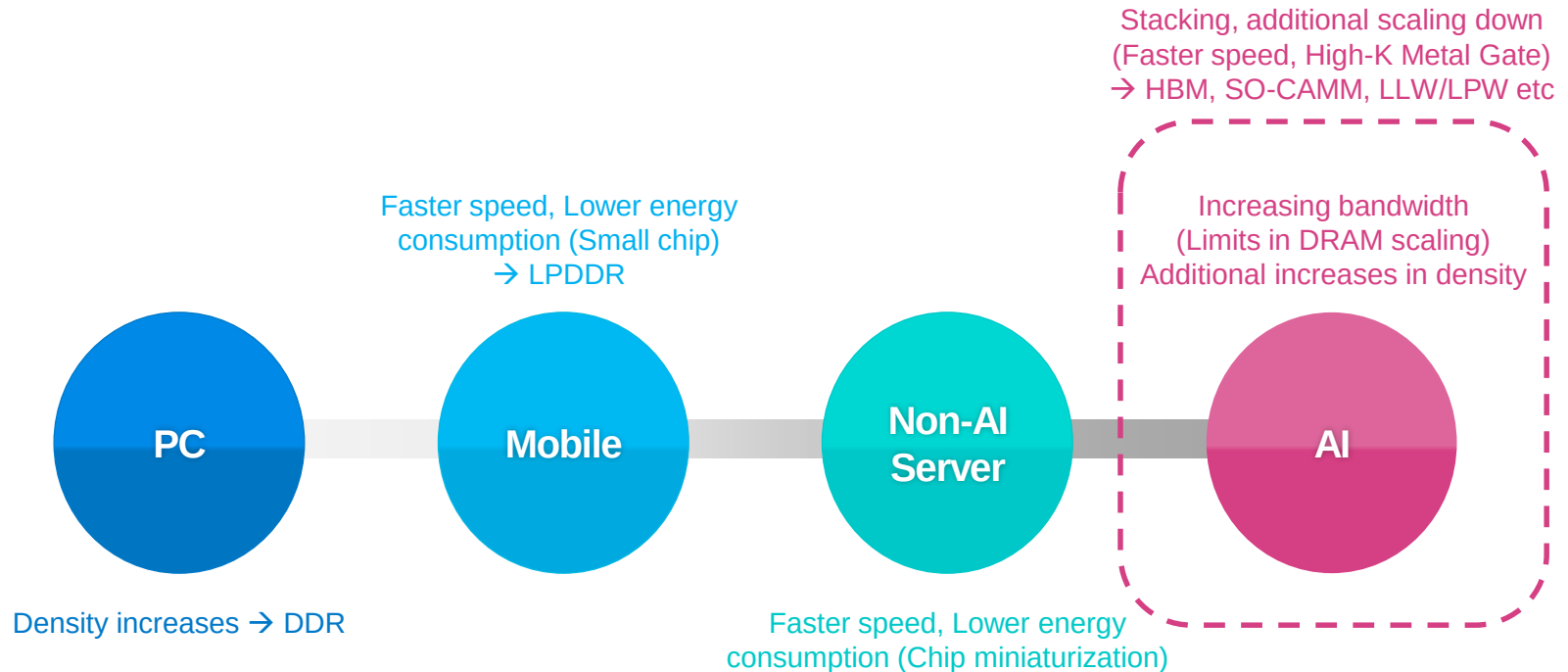
Hybrid Bonding in NAND



- 3D NAND: Divided into the Cell area, which stores actual data, and the Peri area, a CMOS region for driving the Cell.
- Hybrid Bonding combines Cell and Peri from separate wafers. YMTC and Kioxia are in use.

Source: YMTC, Samsung Securities

As times change, technology evolves rapidly



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